Chenghao Ye

Senior Software Engineer | Go | High-Throughput Backend Systems

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Profile

Senior Software Engineer specialising in Go microservices; **MSc in Information Technology**, Carnegie Mellon University.

Five years' experience **building and operating Go backend services** for Tencent Music's platform with **over 500 million monthly active users**, focusing on API design, data pipelines, transaction workflows and observability. Proven track record improving latency, storage efficiency and production operability, from an internal segmentation platform **handling 1.9 billion queries per day** to a music-promotion order platform with annual transaction value exceeding **RMB 100 million**.

Professional Experience

Tencent Music Entertainment -- Commercial Advertising — **Apr 2021 -- Mar 2026**

Shenzhen, China | Promoted from Software Development Engineer I to II in Jan 2023 and to III in Jan 2025.

User Segmentation Platform

Solution Designer and Primary Developer | Greenfield shared platform for targeting, entitlements and A/B testing.

- Designed and primarily developed a shared internal segmentation platform in Go, Redis-compatible storage and Kafka, providing a unified API for targeting, entitlements and A/B testing; adopted by **57 downstream services** and handling **1.9 billion queries per day**, with peaks of approximately **60,000 QPS**.
- Replaced one-key-per-user-segment-relationship storage with compact per-user segment lists; the total production storage footprint fell from approximately **600 GB to 90 GB**, lowering storage cost. Migrated through parallel imports and query fallback to the old system before retiring it after stability checks.
- Built a rate-limited Kafka/Lua import path with slot-based batching, deduplication and persisted progress for resumable imports; reduced import time for the same **30 million records** from approximately **726 seconds with Pipeline to 101 seconds** in a benchmark.
- Combined Maglev consistent hashing for request affinity with BigCache, achieving a **75% local-cache hit rate**.

Music Promotion Platform

Order Module Lead | RMB 100M+ annual transaction value; five-figure query QPS during campaigns; 100K+ in-flight orders.

- Led the evolution of order fulfilment for fans, artists and labels, separating individual orders from aggregated delivery tasks; connected wallet top-ups, pricing, payment pre-authorisation, daily delivery plans, result collection and settlement reconciliation.
- Designed and built aggregation by song for eligible small paid promotion orders to address task fragmentation and over- or under-delivery; preserved individual payment, progress and settlement records and allocated delivered play counts back to each order. Within roughly **two months of launch**, approximately **850,000** orders were aggregated, reducing delivery tasks by **40% overall** and **91%** in one major campaign.
- Used Redis caching, batch reads and asynchronous processing for non-critical side effects, reducing average order-placement latency from **210 ms** to **approximately 90 ms**.

Advertising Platform

Core Contributor | Shared platform serving 17+ media applications; **16.4B internal requests/day**.

- Drove adoption of a shared reward callback protocol, centralised reward configuration and built self-service integration tools; supported **10+ systems and six protocol families**, moving most new-media onboarding from weeks to days. An associated experiment in the K-song karaoke app recorded a **1.8% revenue uplift**.
- Built backend comparison of device-cached and live ads with validation, candidate normalisation, ranking and eCPM fallback. The best experiment variant raised revenue per user at the tested placement by **17.4%** and overall revenue in QQ Music's ad-supported free mode by **2.45%**.
- With the monitoring team, cut ad-serving entry-layer scenarios from **100K+ to about 600** and attributes from **10K+ to 5,500**, retaining coverage; alerted immediately on urgent issues, reported lower-priority ones next day, and ran old and new alerts in parallel.

Technical Skills

- **Programming & Scripting:** Go / Golang (advanced); Java, Python, Bash & Shell Scripting (familiar)
- **Backend & Data Technologies:** Redis, Kafka; MySQL, MongoDB, Elasticsearch; REST APIs and Protobuf-based RPC
- **Observability & Engineering:** Production monitoring with Prometheus and Grafana; performance analysis, reliability reviews and incident response
- **Systems Architecture:** Microservices, multi-tenancy, event-driven systems, horizontal scaling; caching, consistent hashing, transactions, data consistency, idempotency, rate limiting and resilience

Education

- **Carnegie Mellon University** — **2019 -- 2020**
  Master of Science in Information Technology — DISTINCTION (GPA 3.8)
  Information Security and Privacy Policy, Distributed Systems, API Design and Implementation, Software Construction.
- **The University of Edinburgh** — **2017 -- 2019**
  Bachelor of Science in Computer Science — FIRST CLASS HONOURS (GPA 4.0)
  Internet of Things (IoT), Computer Security, Database Systems, Machine Learning & AI, Software System Design.

Publications

"Retrofitting Security and Privacy Measures to Smart Home Devices" First author, 6th IEEE International Conference on Internet of Things: Systems, Management and Security, 2019.